Theoretical Study on CO_2 Fixation Catalyzed by Polyoxometalate

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1. Introduction

Polyoxometalates (POMs) are anionic clusters formed by condensation of octahedral oxides of early-transition metals at edges or vertices. Lindqvist-type POM of Ta has recently been synthesized in the liquid phase [1]. According to the elemental analysis, this POM is coordinated by protons and tetrabutylammonium cations (TBAs) to be $TBA_{4.3}H_{3.7}Ta_6O_{19}$. It exhibits high catalytic activity for CO_2 fixation, but its activation mechanism remains unclear. In this study, we aim to determine a reaction path of the CO_2 fixation into styrene oxide (SO) catalyzed by the Ta-based POM, thereby unveiling why the POM lowers the reaction barrier.

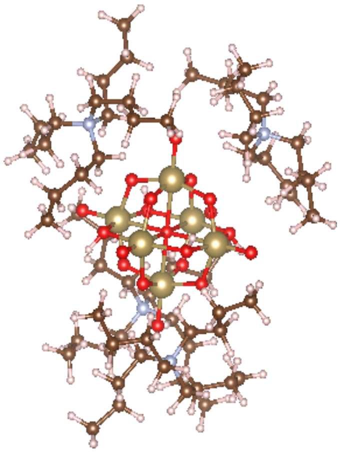

Figure 1. The optimized structure of Ta6.

2. Method and Results

We constructed a model cluster of $TBA_4H_4Ta_6O_{19}$ (Ta6) with the highest symmetry (S_4), including the Ta_6O_{19} core with O_h symmetry. First, we obtained its stable structure using the DFT calculation at the B3LYP/3-21G (H, C, N, O) and LanL2DZ (Ta) level of theory. Subsequently, we set the orientation of the reactants so that the overlap between the HOMO of Ta6 and the LUMO of SO is significant and conducted the transition-state search. The obtained reaction path consists of two steps. The first one is the ring opening of epoxide, and the second one is the ring closing with CO_2 incorporated. The rate-determining step is the former, and its reaction barrier (92.9 kJ/mol) is less than half of that without any catalyst (200 kJ/mol). In the symposium, we will also discuss the reason for lowering the reaction barrier from the viewpoint of electronic structures.

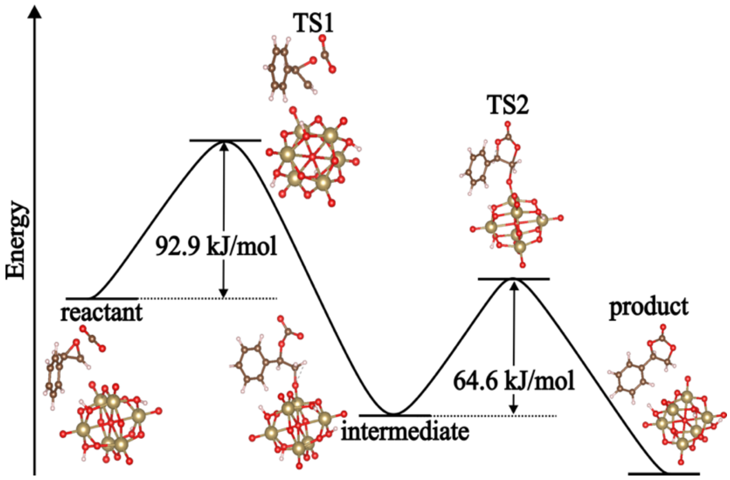

Figure 2. The reaction profile of CO_2 fixation catalyzed by Ta6. TBAs are not displayed for clarity.

References [1] S. Hayashi *et al.*, *J. Phys. Chem. C*, **122**, 29398-29404 (2018).